

VISHAK SHASHIKUMAR

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Education

Northeastern University

Master of Science in Applied Machine Intelligence

Boston, MA, USA

December 2026

JSS Science & Technology University

Bachelor of Engineering in Computer Science

Mysuru, India

May 2021

Technical Skills

Languages: Python, SQL, TypeScript/JavaScript, R, Bash

AI & ML: Large Language Models (Claude, OpenAI, Mistral), Retrieval-Augmented Generation (RAG), prompt engineering, prompt-injection hardening, agent tracing and evaluation (LLM-as-judge), embeddings and vector search, NLP, time-series forecasting (SARIMAX), anomaly detection

Frameworks/Libraries: FastAPI, Flask, Next.js/React, Pandas, NumPy, scikit-learn, PyTorch, Sentence Transformers, LangChain, SQLAlchemy 2.x, Pydantic, pytest, Streamlit, Matplotlib

Data & Analytics: PostgreSQL, MySQL, DynamoDB, Vector DBs (FAISS), ETL/ELT pipelines, data modeling, statistical analysis, Tableau, Power BI, KPI dashboards, trend and cohort analysis

Tools & Cloud: Git/GitHub, Docker, Kubernetes, Ollama, AWS (S3, SageMaker, DynamoDB), CI workflows, Jupyter

Professional Experience

Ipsier Labs

AI Engineer Intern

Remote

January 2026 – July 2026

- Built and secured FastAPI services powering the LLM features of a consumer application, adding vector-database retrieval (RAG) to ground model responses in real application data
- Hardened LLM-facing endpoints against prompt injection with input sanitization and prompt isolation, validated by an automated pytest suite of adversarial payloads
- Delivered end-to-end LLM cost observability: per-call token instrumentation, persistence to an indexed usage table, an authenticated endpoint aggregating spend by user and operation, and daily-budget alerting
- Eliminated silent data loss in that pipeline by replacing fire-and-forget async writes with awaited transactional commits, so every model call was recorded before the request returned

Balaji AutoTech Pvt Ltd

Data Analyst

Mysuru, India

January 2022 – June 2024

- Engineered scalable ETL pipelines to extract, clean, and unify manufacturing data, delivering 95%+ accuracy across analytics workflows
- Developed anomaly detection models and interactive Tableau and Power BI dashboards that cut equipment downtime 15% through real-time fault monitoring
- Deployed containerized analytics services with Docker and Kubernetes, giving teams reliable API access to aggregated metrics
- Partnered with cross-functional engineering teams to optimize data flow, improving pipeline performance and visibility

Projects

AgentLens — LLM Agent Profiling & Optimization Framework

July 2026 – August 2026

- Built a framework that audits other LLM agents: span-based tracing of every tool and model call, a golden-task eval harness pairing rule checks with LLM-as-judge scoring, and a taxonomy of four failure patterns ranked by cost
- Closed the loop with an optimizer that rewrites the agent's system prompt from the diagnosis and re-runs the benchmark to verify it, cutting token spend 44–52% and tool calls 25–45% with no drop in task success rate

Second Brain for Life — Personal AI Knowledge & Journaling Platform

May 2026 – July 2026

- Built a fully local RAG system over a personal knowledge base: FAISS search across sentence-transformer embeddings feeds a self-hosted Mistral model (Ollama), streamed token-by-token from FastAPI with no data leaving the machine
- Modeled a normalized PostgreSQL schema (typed SQLAlchemy 2.x, Pydantic-validated REST) serving aggregation queries to a TypeScript/Next.js dashboard (React, Recharts) for journaling and retrieval trends

CareCast — Pediatric ICU Forecasting System

October 2025 – December 2025

- Built a SARIMAX forecasting pipeline on large-scale HHS hospital data predicting 7–60-day pediatric ICU utilization, with automated cleaning and capacity-shock simulations
- Shipped an interactive Streamlit platform with KPI cards, sparkline trends, and state-level ICU maps, open-sourced under the MIT License